CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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PROLOG PROGRAM TO SUM INTEGERS FROM 1 TO N

Aim:

To write a Prolog program to calculate the sum of integers from 1 to N.

Objective:

To implement recursive addition in Prolog using facts, rules and arithmetic operations.

Algorithm:

1. Start.
2. Read the value of N.
3. If N = 0, return sum as 0.
4. Otherwise, recursively calculate the sum from 1 to N-1.
5. Add N to the recursive result.
6. Display the sum.
7. Stop.

Flowchart:

```
START
  |
  v
Read N
  |
  v
Is N = 0?
 /      \
Yes      No
 |        |
Sum=0   Sum(N-1)
 |        |
 |      Add N
 |        |
  \      /
   v    v
 Display Sum
     |
     v
    STOP
```

Prolog Program:

```
% Sum of integers from 1 to N


sum_to_n(0, 0).


sum_to_n(N, Sum) :-
    N > 0,
    N1 is N - 1,
    sum_to_n(N1, S1),
    Sum is N + S1.
```

Output:

```
54 ?- sum_to_n(10, S).
S = 55
```

Result:

Thus, the Prolog program successfully calculates the sum of integers from 1 to N.